

GABLS1: measured scheme-native WENO flux

GABLS1 scheme-native WENO transport | 22 September 2026. Matched 12.5 m, 32³, 400 m, WENO9, one face, 300 s filter, factor one, seed 123, nine-hour LES. The old admitted factor-one covariance run is the control.

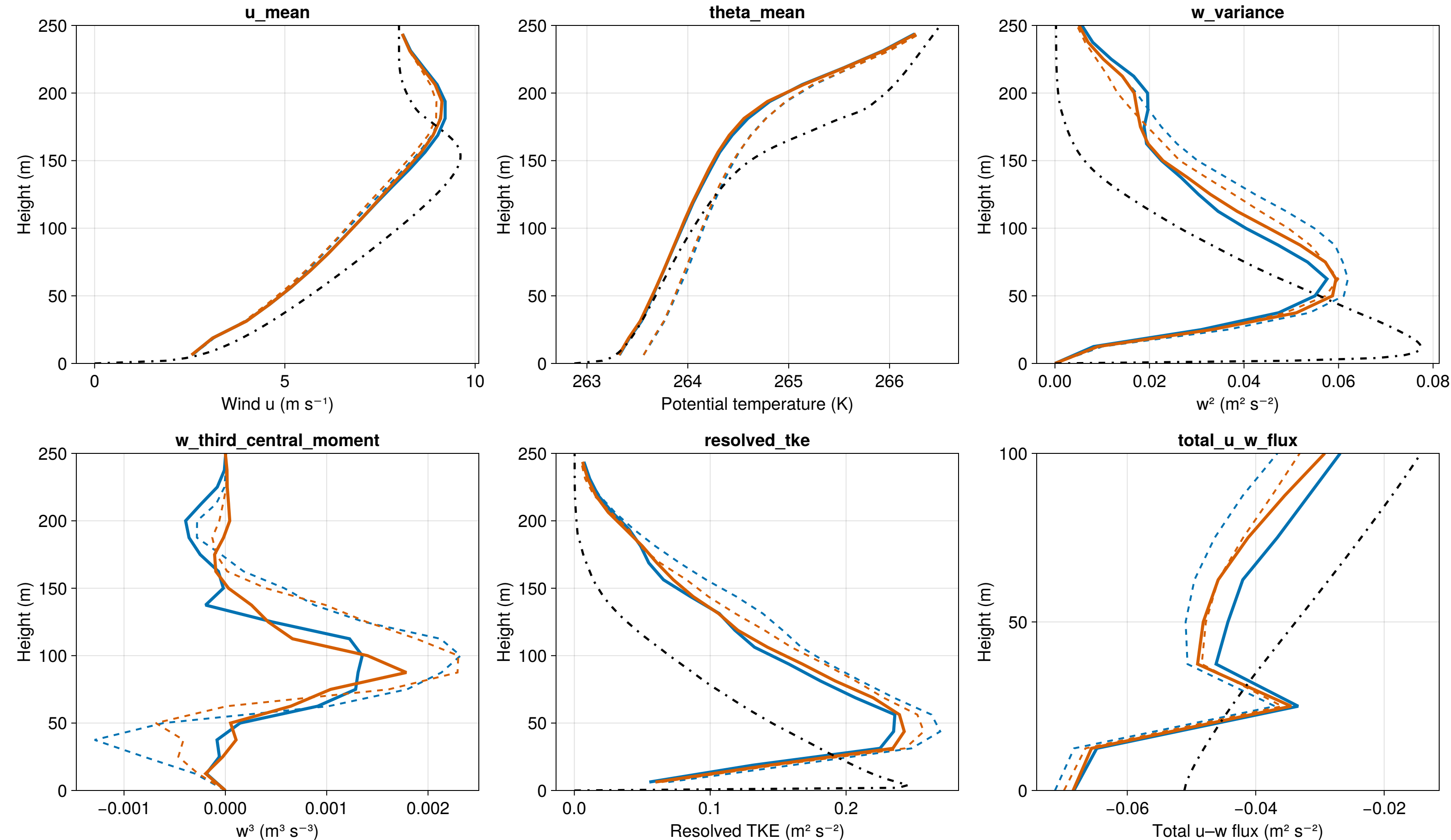
At the first interior face, the measured 8–9 h WENO numerical correction to u-momentum flux is $-0.000228 \text{ m}^2/\text{s}^2$, versus covariance $-0.007749 \text{ m}^2/\text{s}^2$ (2.95%). The heat correction is $1.4\text{e-}10 \text{ K m/s}$, effectively zero versus covariance -0.001472 K m/s . The saved covariance plus correction equals reconstructed flux at both diagnostic faces for momentum and scalars. These measured corrections are far below the assumed extra flux behind factors two and three in this case.

Mean 8–9 h first-face momentum viscosity falls 1.3214 to 1.2958 m^2/s (-1.9%); heat diffusivity rises 1.2160 to 1.2642 m^2/s (+4.0%). Integrated resolved TKE rises 3.8%, peak w^2 rises 3.1%, diagnosed boundary-layer height falls 7.4 m. One evolving realization cannot distinguish a systematic response from trajectory divergence. The profiles remain near the covariance control and show substantial coarse-run departures from the fixed 1 m LES median in resolved turbulence.

Source and data audit: immutable Breeze/Evaluation manifest SHA-256 72f1bc03 and 792 source-file hashes passed; paired initial-theta SHA-256 1f5db33f matched. H100 gate 7496 passed; science 7497 had durable exit zero and CASE_DONE at 32400 s. Raw output passed exact 1/19/541 initial/profile/series schedules, finite values, native 32/33-level heights, and flux decomposition on both faces. Figures use Julia. The reference is an LES median, not observations. Historical factor and control outputs were not rerun.

GABLS1: scheme-native transport versus covariance

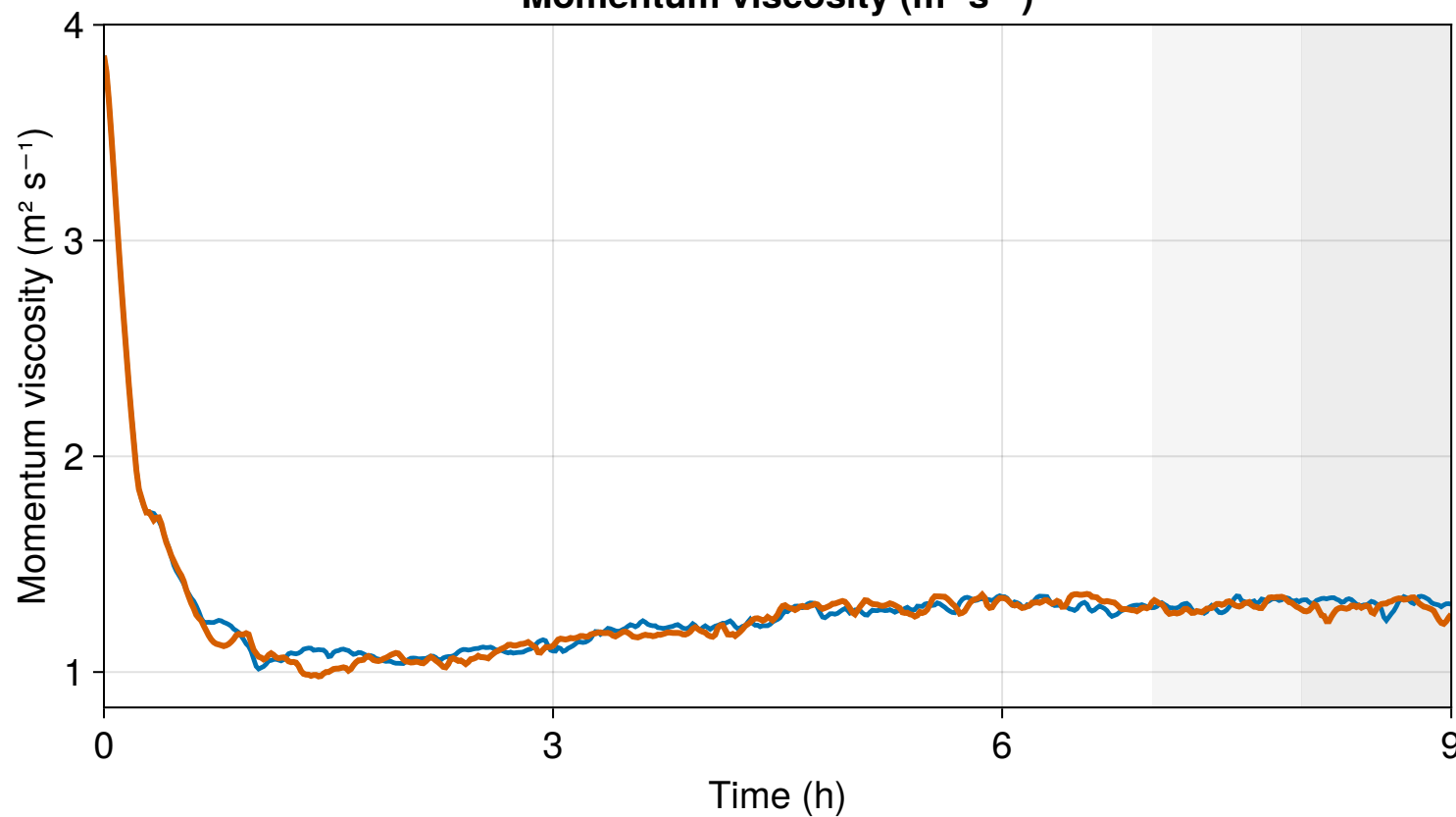
12.5 m · WENO9 · one face · 300 s · factor 1 · same seed; dashed 7–8 h, solid 8–9 h



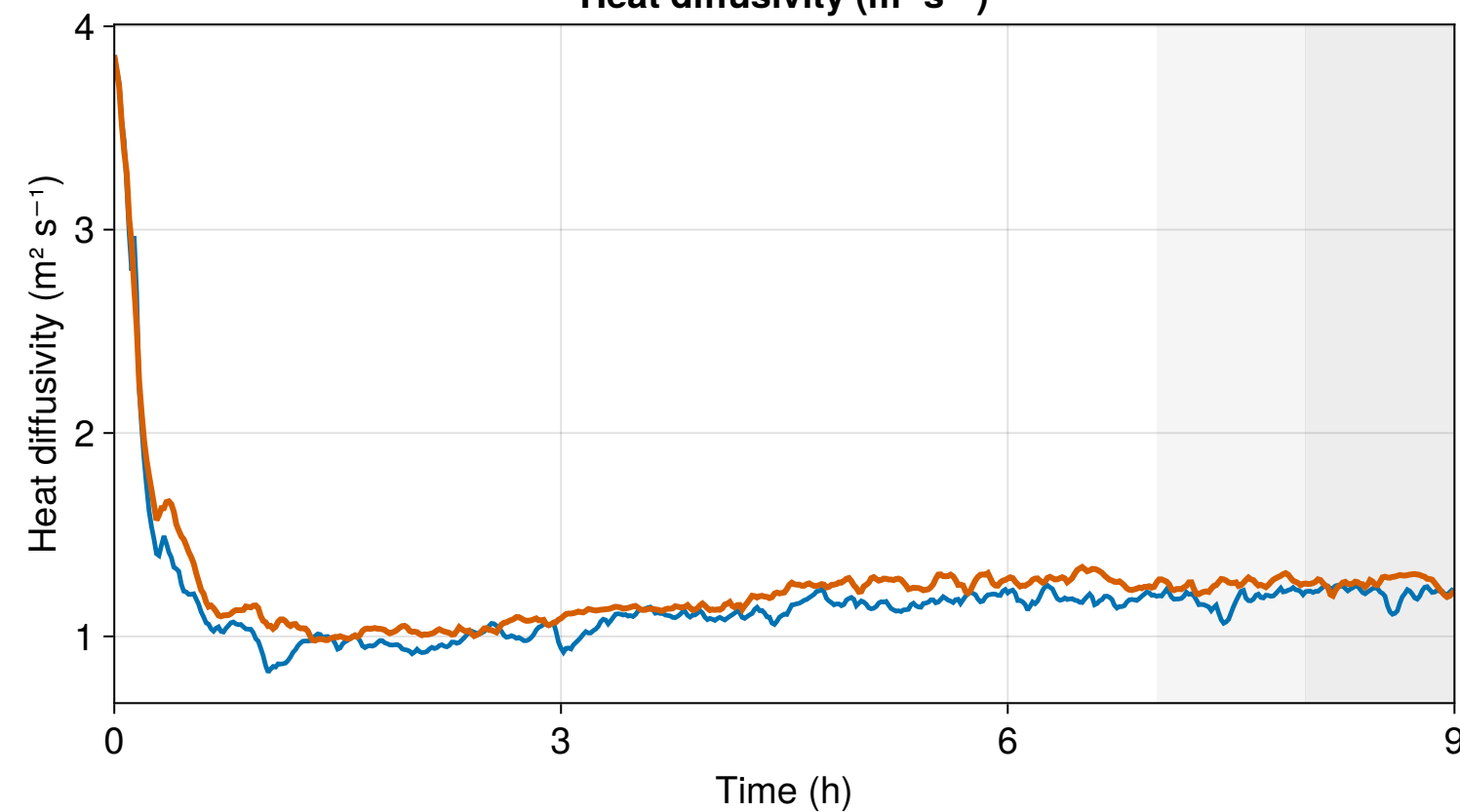
What the transport estimator changes at the first interior face

Same nine-hour forcing and discretization · 12.5 m supported face · shaded final two hours

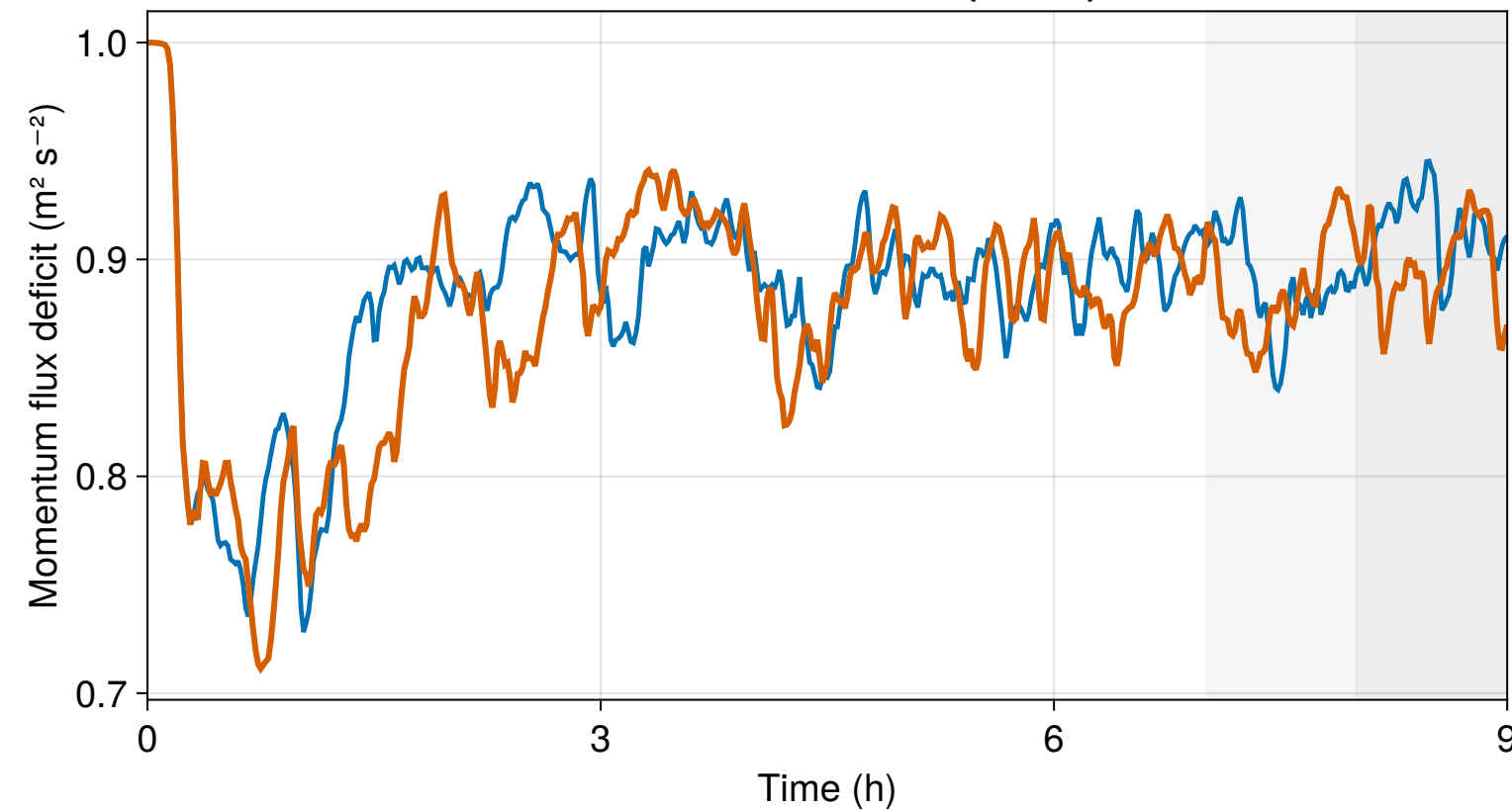
Momentum viscosity ($\text{m}^2 \text{s}^{-1}$)



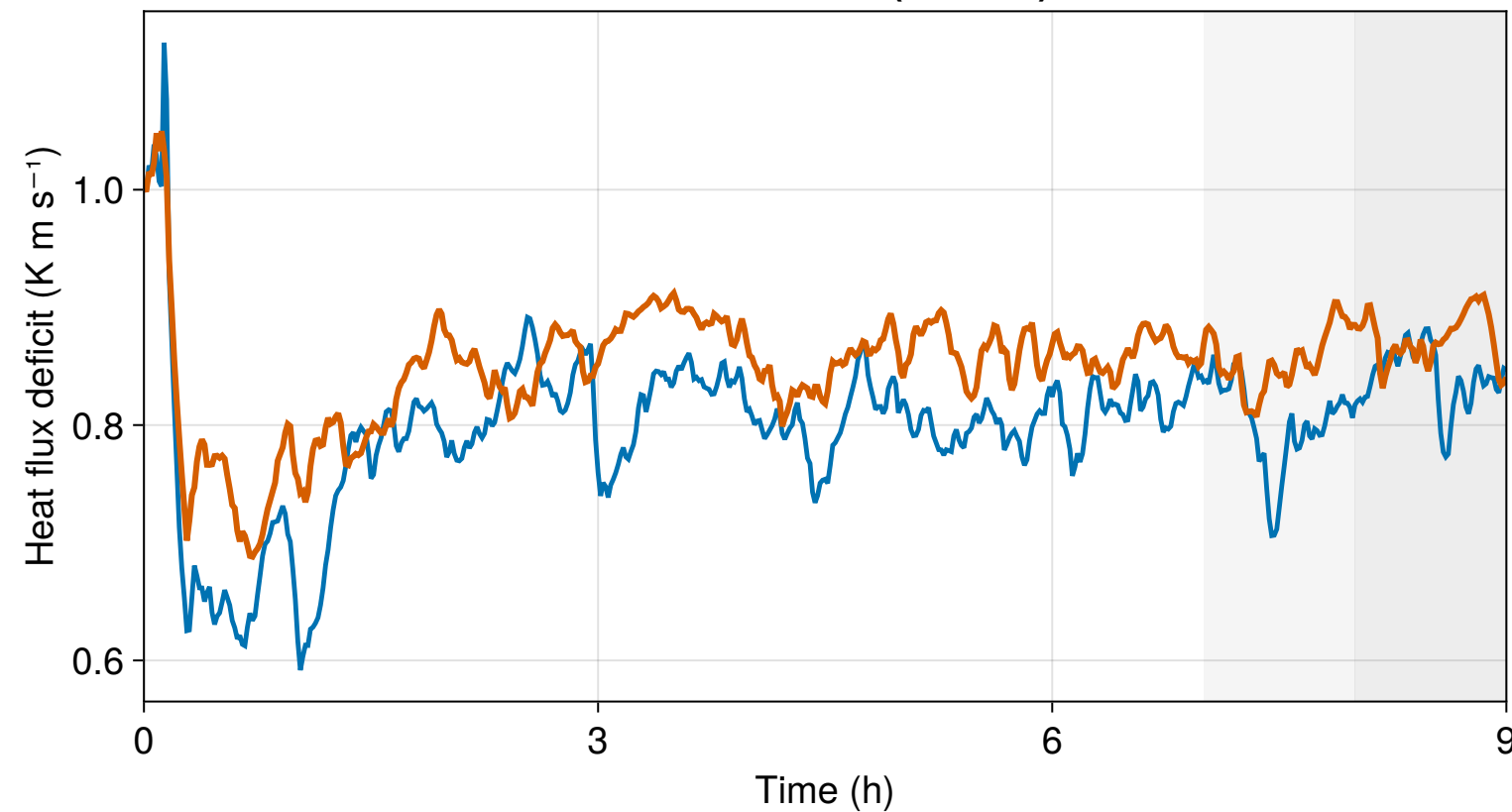
Heat diffusivity ($\text{m}^2 \text{s}^{-1}$)



Momentum flux deficit ($\text{m}^2 \text{s}^{-2}$)



Heat flux deficit (K m s^{-1})

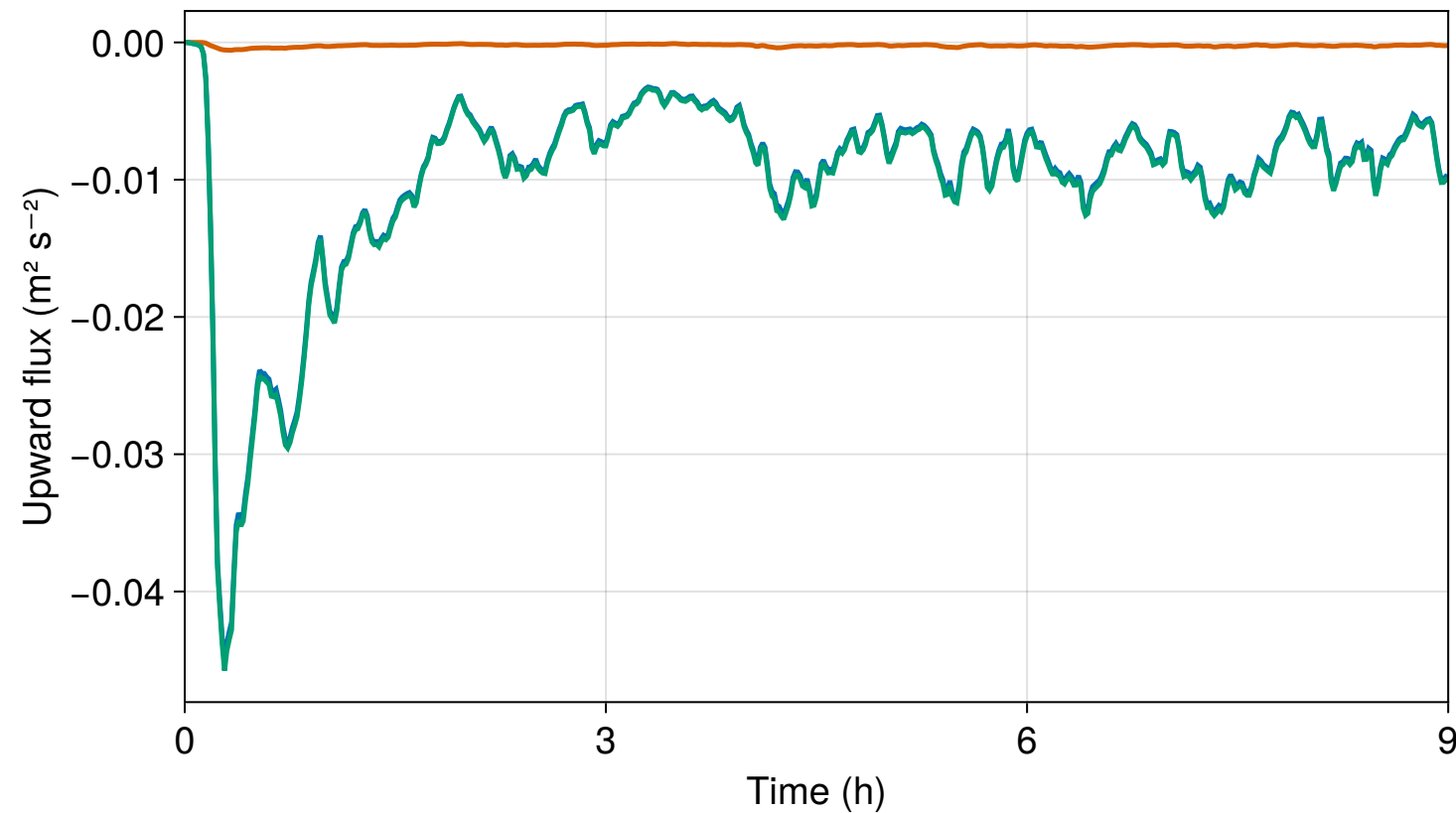


— Covariance control — Scheme-native

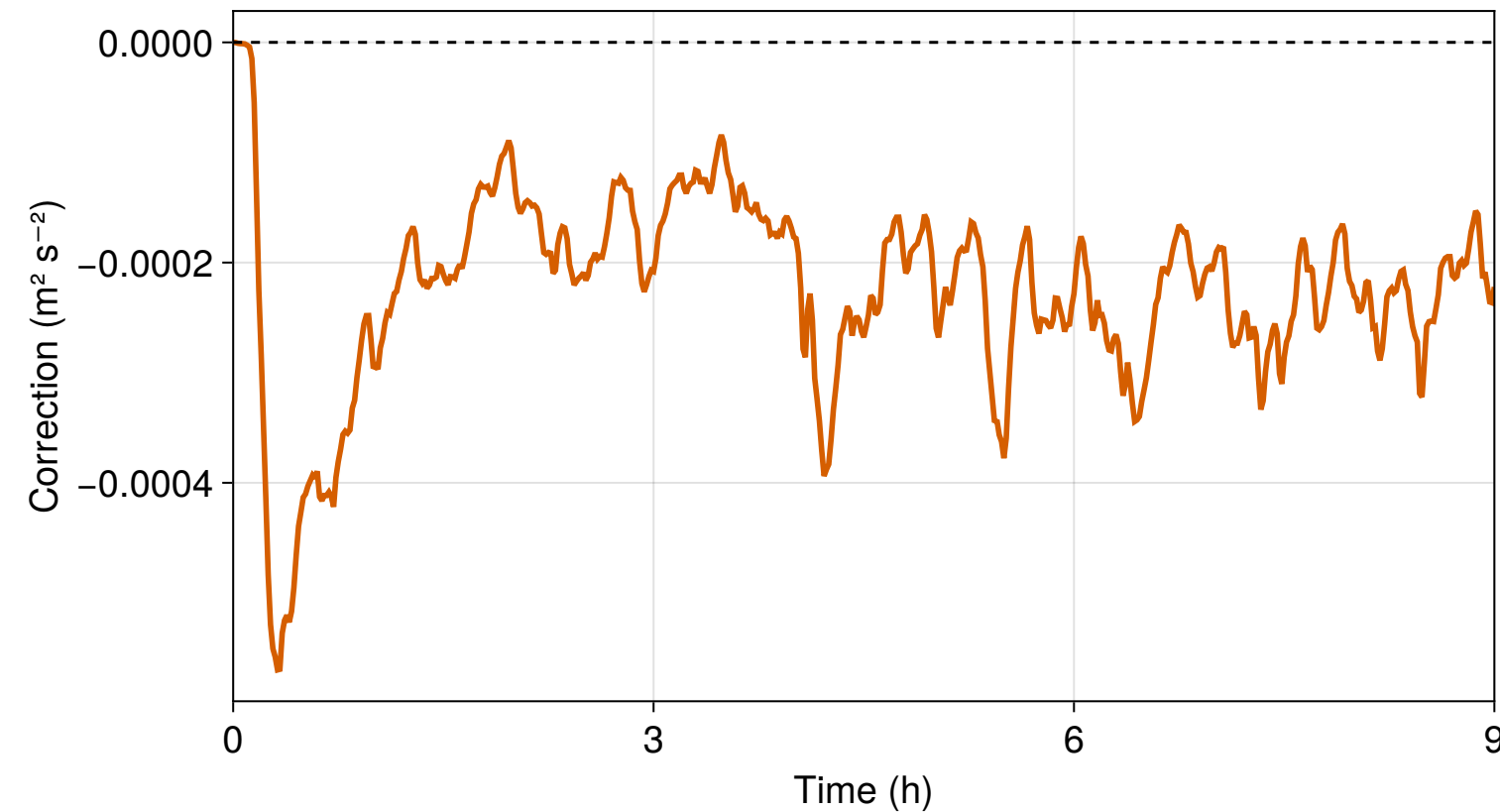
Measured WENO transport correction

Scheme-native run only · covariance + numerical correction = reconstructed flux

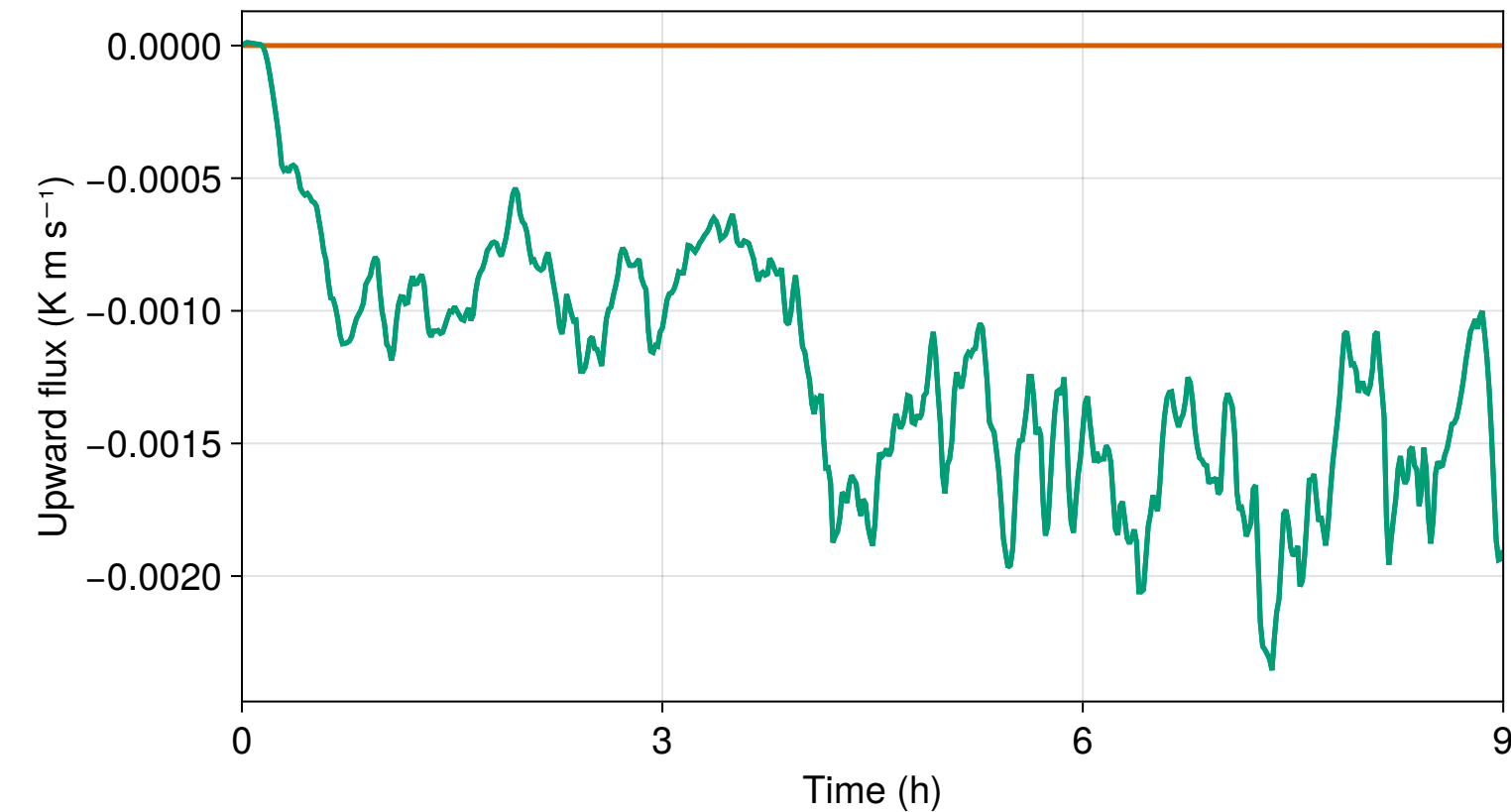
Momentum flux



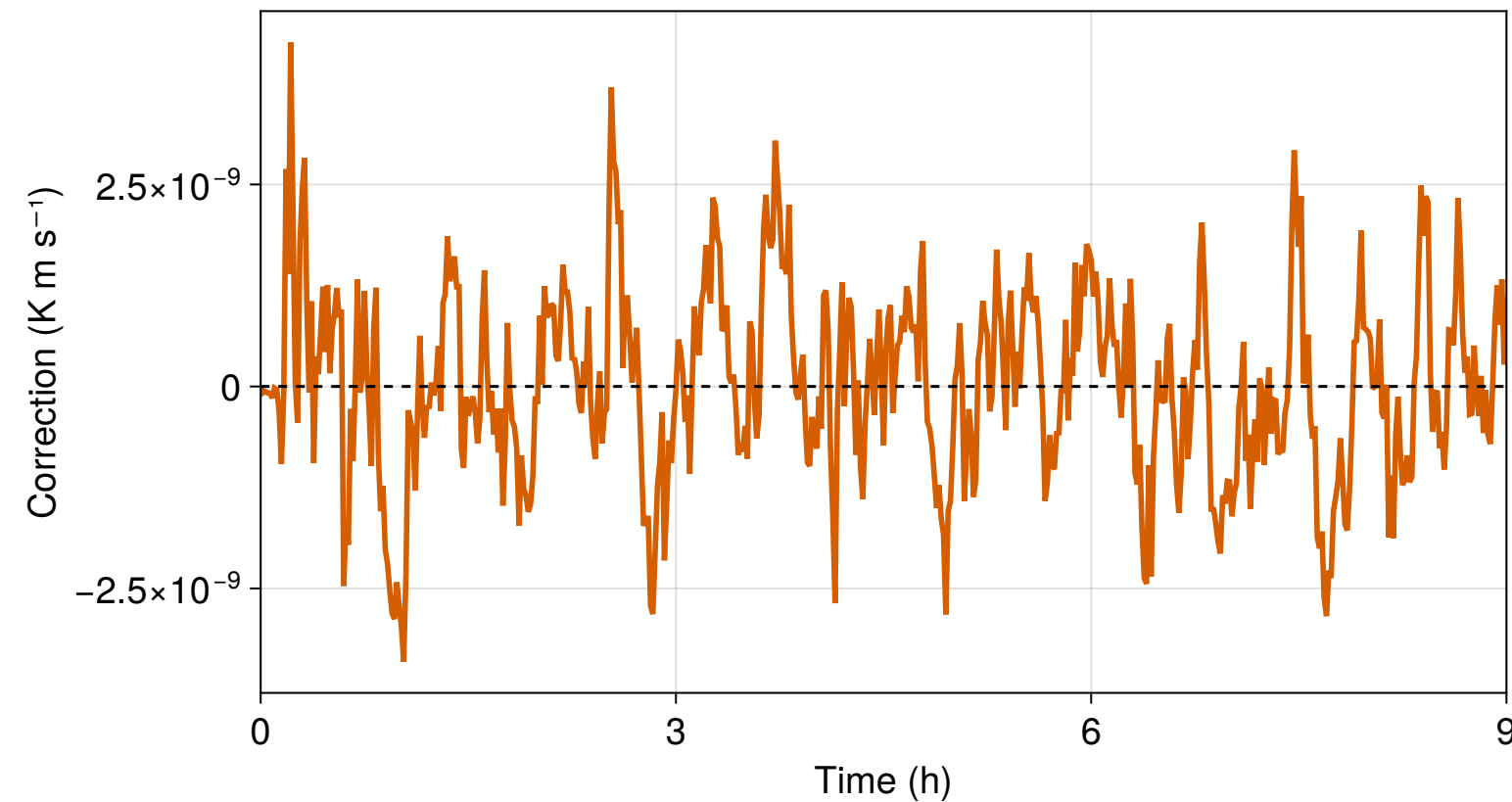
Momentum numerical correction



Heat flux



Heat numerical correction



— Covariance — Numerical correction — Reconstructed